ADAM : Freeing Appearance in Frequency-Aware Image Editing with Amplitude Manipulation

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Introduction

Motivation : Frequency-aware Image Editing

- **Text-Guided Image Editing**
  - Preserving the source image's structural layout while injecting new semantic attributes from target prompt
- **The trade-off : How to balance both?**
  - Approach 1: Attention ManipulationMethods
    - Pixel-level operations via attention map manipulation
    - Limited versatility and generalization
  - Why Frequency-based Editing?
    - Pixel-level attention entangles structure and semantics
    - Frequency domain separates **global structure** from **local appearance**
    - Enables controlled editing with structural preservation

Limitation of Prior Works

- **Approach 2: Frequency-Based Editing**
  - Prior frequency-based methods anchor low frequencies to unconditional guidance, which leads to "locking" the style of the source image.
  - Fails to reflect significant appearance-level modifications requested by the target prompt
- **Our Key Idea**
  - Low-frequency components should not be entirely assigned to unconditional guidance
  - Low-frequency components should **partially follow conditional noise**
  - Decoupling **amplitude (appearance)** and **phase (structure)** is needed for frequency-aware control

Proposed Method

Preliminaries: Amplitude and Phase in Diffusion Noise

- Given conditional, unconditional noise predictions $\epsilon_c, \epsilon_{uc}$ from a U-Net, we transform these spatial signals into the frequency domain using the 2D Fast Fourier Transform (FFT) : $\boldsymbol{X_c} = \mathcal{F}(\boldsymbol{\epsilon_c}), \quad \boldsymbol{X_{uc}} = \mathcal{F}(\boldsymbol{\epsilon_{uc}})$
- A frequency spectrum can be decomposed into its **amplitude** $\mathcal{A}$ and **phase** Φ : $\boldsymbol{X(u,v)} = \boldsymbol{\mathcal{A}(u,v)} \cdot \boldsymbol{e^{i\Phi(u,v)}}$
  - Amplitude = $\mathcal{A}(u,v) = |X(u,v)|$: related to **appearance** (color, brightness, global style)
  - Phase = $\Phi(u,v) = \angle X(u,v)$: related to **structural information** (edges, shapes, and spatial layout)

ADAM : Appearance-aware Decoupled Amplitude Manipulation

Three-Band Spectral Partitioning

Using two radial distance thresholds D_{min} , D_{max} we partition the frequency domain

- **Low–frequency Band** $\boldsymbol{M_L}$: $D \le D_{min}$
  : global appearance & coarse layout
- **Mid-frequency band** $\boldsymbol{M_H}$: $D_{min} < D \le D_{max}$
  : object boundaries & structural integrity
- **High–frequency band** $\boldsymbol{M_H}$: $D > D_{max}$
  : fine textures & local semantic details

Amplitude Emphasized Guidance

$$\boldsymbol{X} = \boldsymbol{\mathcal{A}(X_c)} \cdot \boldsymbol{e^{i\Phi(X_{uc})}}$$

Instead of selecting either X_c or X_{uc} monolithically, we decouple and synthesize a new spectrum

- To **inject the target appearance** , we enforce the amplitude of the conditional prediction $\mathcal{A}_c$
- To **anchor the global structure** to the source image, we retain the phase of the unconditional prediction Φ_{uc}

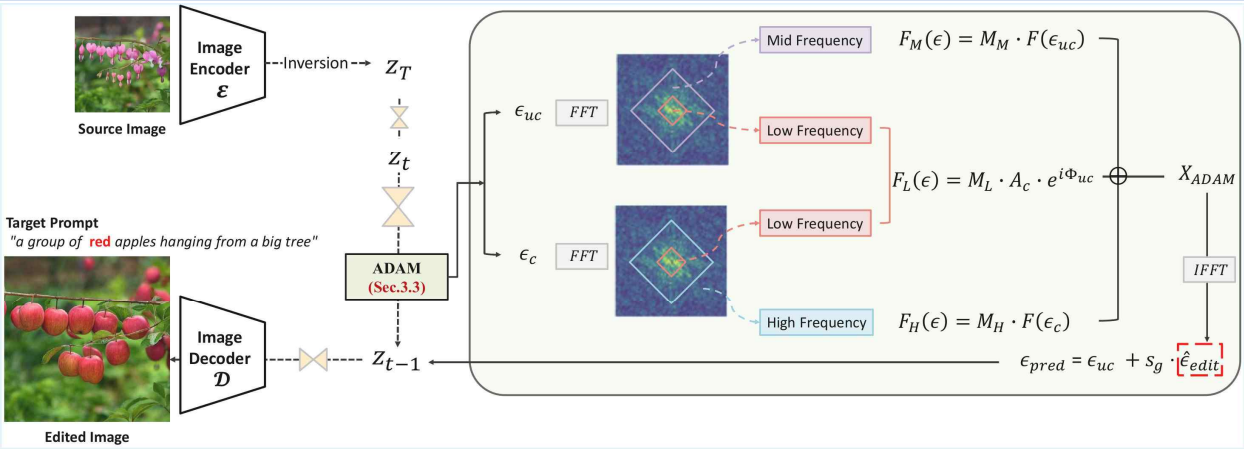
Integrated Guidance Modulation

$$X_{ADAM}(u,v) = \begin{cases} \mathcal{A}(X_c) \cdot e^{i\Phi(X_{uc})} & \text{if } (u,v) \in M_L \\ X_{uc}(u,v) & \text{if } (u,v) \in M_M \\ X_c(u,v) & \text{if } (u,v) \in M_H \end{cases}$$

- The final edited noise spectrum X_{ADAM}
  : amplitude emphasized guidance at low band

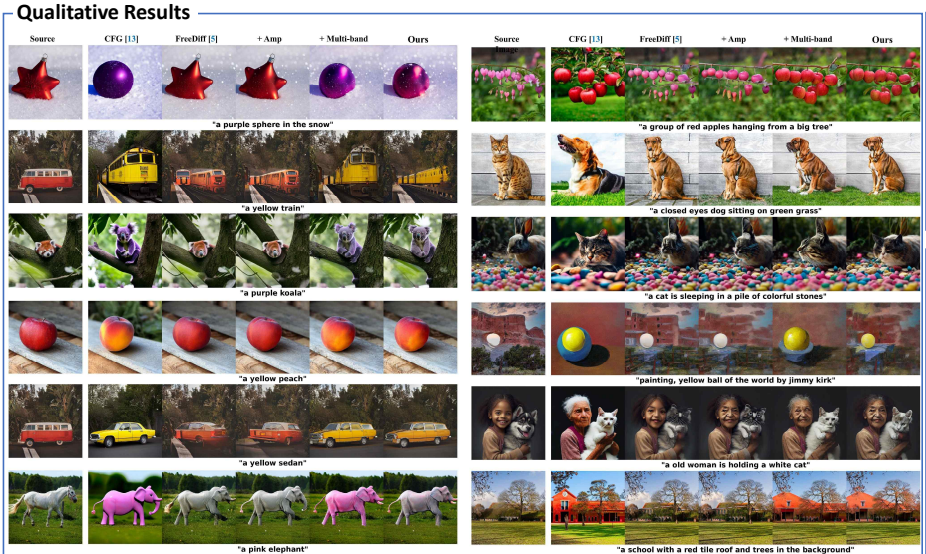
The final edited noise prediction

$$\widehat{\epsilon_{edit}} = \mathcal{F}^{-1}(X_{ADAM})$$



Experimental Results

Qualitative Results



(a) Qualitative comparison results on the Appearance Test Set (b) Qualitative comparison results on the PIE Dataset [14]

Datasets & Evaluation metrics

- **Datasets**
  1. Appearance Split
     : Representative test cases curated from LoMOE and OIR dataset
  2. PIE-Bench
- **Metrics**
  1. CLIP Score ↑
  2. LPIPS ↓

Quantitative Results

Method	CLIP Score (↑)	LPIPS ↓
CFG [13]	0.353 ± 0.042	0.466 ± 0.104
FreeDiff [5]	0.254 ± 0.028	**0.188 ± 0.077**
+ AMP	0.270 ± 0.023	0.210 ± 0.071
+ Multi-band	**0.350 ± 0.034**	0.309 ± 0.111
ADAM (Ours)	**0.331 ± 0.039**	**0.290 ± 0.113**

Appearance Split

Method	CLIP Score (↑)	LPIPS ↓
CFG [13]	0.351 ± 0.039	0.517 ± 0.151
FreeDiff [5]	0.281 ± 0.058	**0.213 ± 0.120**
+ AMP	0.289 ± 0.057	0.245 ± 0.123
+ Multi-band	**0.327 ± 0.051**	0.354 ± 0.133
ADAM (Ours)	**0.313 ± 0.051**	**0.321 ± 0.124**

PIE-Bench